

SURIN, Thailand

WMO: 484320

Lat: 14.874N

Lon: 103.499E

Elev: 483

StdP: 14.44

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	59.6	62.2	49.8	54.0	66.9	52.4	59.5	68.7	15.7	76.4	13.8	76.8	2.7	30	0.319

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	17.6	99.5	77.3	97.7	77.3	95.9	77.3	80.8	91.0	80.3	90.1	79.9	89.4	4.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.5	150.8	83.9	78.1	148.6	83.6	77.6	146.1	83.2	44.9	91.1	44.4	90.5	43.9	89.8	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
11.9	10.0	8.8	DB	55.0	101.9	2.6	1.5	53.1	103.0	51.6	103.9	50.2	104.7	48.3	105.8
			WB	51.5	82.4	2.8	1.1	49.4	83.2	47.8	83.8	46.2	84.4	44.1	85.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.0	76.6	80.4	84.5	86.7	85.6	84.7	83.4	83.0	82.1	81.3	79.2	76.1
	DBStd	4.78	4.61	5.13	4.40	3.48	2.82	2.26	2.20	2.00	2.10	2.32	3.80	4.78
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	1	0	0	0	0	0	0	0	0	0	0	0	1
	CDD50	11667	826	851	1071	1100	1103	1041	1035	1023	964	969	876	808
	CDD65	6193	361	431	606	650	638	591	570	558	514	504	426	344
	CDH74	68386	3427	4846	7657	8418	7729	6877	6218	5945	5207	4981	3994	3087
	CDH80	29840	1401	2355	4104	4558	3689	2984	2401	2201	1758	1721	1524	1144
Wind	WSAvg	2.6	2.5	2.5	2.5	2.4	2.4	2.8	2.9	2.7	1.9	2.4	3.0	3.4
Precipitation	PrecAvg	50.7	0.1	0.4	0.9	3.3	6.8	6.5	7.3	8.2	11.2	5.0	1.0	0.1
	PrecMax	64.0	0.6	2.5	2.0	8.9	17.8	11.6	13.5	14.5	20.4	12.6	3.2	1.1
	PrecMin	38.8	0.0	0.0	0.0	0.2	1.8	3.6	3.3	2.6	5.2	1.4	0.0	0.0
	PrecStd	7.1	0.2	0.6	0.6	2.4	4.2	2.4	2.7	3.3	4.7	3.1	1.0	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.9	98.0	100.9	102.5	100.3	96.7	95.2	93.5	92.6	92.0	93.6	93.1
		MCWB	73.6	75.1	75.9	77.6	79.5	79.0	79.0	79.4	79.6	78.3	77.1	74.6
	2%	DB	91.9	95.9	98.8	100.1	97.3	94.6	92.7	91.7	90.8	90.3	91.6	90.8
		MCWB	73.3	74.9	76.0	77.7	79.2	79.0	78.9	79.1	79.3	77.7	76.3	74.1
	5%	DB	90.0	94.1	97.0	98.0	95.1	92.8	91.0	90.3	89.3	89.0	89.8	88.8
		MCWB	72.8	74.6	76.1	77.9	79.1	79.1	78.7	78.9	78.9	77.3	75.7	73.3
	10%	DB	87.6	91.7	95.1	95.7	92.9	91.0	89.4	88.8	87.8	87.6	87.9	86.4
		MCWB	71.9	74.4	76.2	77.8	79.0	78.9	78.5	78.6	78.6	77.1	74.7	72.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.3	78.6	79.7	80.9	81.6	81.1	80.8	80.9	81.0	80.3	79.2	77.5
		MCDB	88.8	92.1	94.5	95.4	93.2	91.2	89.9	89.2	88.6	87.6	88.3	88.0
	2%	WB	75.9	77.5	78.6	80.0	80.7	80.4	80.0	80.1	80.1	79.4	78.3	76.1
		MCDB	86.8	90.5	93.1	93.9	92.0	90.4	88.7	88.2	87.5	86.4	87.2	86.6
	5%	WB	74.6	76.7	77.9	79.2	80.1	79.9	79.3	79.7	79.7	78.8	77.4	74.9
		MCDB	85.4	89.3	92.0	92.7	90.7	89.3	87.8	87.5	86.7	85.7	85.5	85.1
	10%	WB	73.3	75.7	77.2	78.7	79.6	79.3	78.9	79.1	79.1	78.2	76.6	73.6
		MCDB	84.5	87.5	90.9	91.8	89.7	88.2	87.2	86.6	85.8	84.5	84.1	84.1
Mean Daily Temperature Range	5% DB	MDBR	21.4	21.5	20.0	17.6	14.8	13.3	12.3	12.0	11.4	12.6	16.3	18.9
		MCDBR	22.2	22.4	21.6	19.7	17.5	15.4	14.5	13.7	13.4	14.0	17.5	19.8
		MCWBR	6.8	5.9	5.2	4.4	4.2	3.8	4.0	4.0	4.2	4.4	5.1	6.0
	5% WB	MCDBR	19.4	19.3	19.2	17.8	15.5	14.1	13.1	12.8	12.5	12.5	15.4	18.0
MCWBR		6.5	5.6	5.3	4.9	4.6	4.3	4.3	4.3	4.5	4.3	4.8	5.9	
Clear-Sky Solar Irradiance	taub	0.498	0.612	0.663	0.600	0.500	0.467	0.460	0.463	0.496	0.523	0.475	0.465	
	taud	2.033	1.758	1.688	1.872	2.191	2.292	2.309	2.292	2.186	2.084	2.198	2.166	
	Ebn at Noon	252	228	221	236	256	262	265	267	259	249	257	258	
	Edh at Noon	53	72	79	66	47	42	42	43	48	51	44	45	
All-Sky Solar Radiation	RadAvg	1590	1665	1705	1823	1803	1701	1613	1555	1485	1567	1571	1537	
	RadStd	85	69	63	73	96	65	103	95	98	103	72	47	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.56	N/A	N/A	N/A	+0.40	+0.22	N/A	N/A	+232
(47) Regional (18 neighbors)		+0.47	N/A	N/A	N/A	+0.45	+0.40	N/A	N/A	+185

Nomenclature: See separate page